



Advanced Physics Laboratories

Welcome!



Advanced Physics Laboratories (APL) 2025–2026

The Advanced Physics Laboratories are located in Beech Hill on the ground and first floors of the B-wing:

- Rooms B001, B002, B003, B004 and B005 (Ground Floor)
- Rooms B103 (Paul Hanratty & Joe Branson's Offices) and B105 (First Floor)

Important: When accessing the labs on the first floor, do so using the stairs at the back of the labs on the ground floor to avoid going through classroom in B106-BH while lectures are taking place!



School of Physics move to Beech Hill

- We are temporarily located in Beech Hill (BH) since 2023. Date to move back to Science TBD.
- BH Building will be closing at **21:30**, so you must leave by then.
- Please let us know if there are any issues with schedules and commuting between UCD main campus and Beech Hill.





School of Physics move to Beech Hill

No Physics Library, common room or dedicated write-up space in Beech Hill.

You will be able to use B106-BH as your base:

- Microwave and kettle available at end of room
- Must make sure that room is kept clean / no strong smells
- At end of day, chairs must be left ready for next day's lectures
- B106 will have lectures scheduled in it – we will post schedule on the door once we have it

Points of contact:

- Main Office (B109–BH)
- Bairbre Fox, School Manager (A109–BH)
- Prof. Dominic Zerulla, Head of School's Office (A108–BH)
- Dr. Ian Mercer, Stage 3 teaching Team coordinator (Science South 0.009)





APL Philosophy

Laboratory Philosophy:

- Educational environment
- Interaction with staff & demonstrators is important
- The key aims of the Advanced Laboratory is to develop independent critical-thinking and transferable skills in an Experimental Physics setting, but with wider applications. **Learn to Think for yourselves!**
- Not only about the report, but also demonstrating understanding of the experiment, including set-up, execution, data analysis, interpretation of results.



APL Philosophy

Laboratory Philosophy:

- Work must be your own– if a report is found to be plagiarised from another student's work, this will be investigated and could result in **both** reports being marked as zero.
- Everything you submit should be your own work, ChatGPT should not be used for the laboratories.
- You must be able to explain and defend everything written in your reports (and code!) during the discussions with staff. Your ability to do so will be part of the grading for the experiment.



Information & Communication

Website for the ADVANCED PHYSICS LABORATORY

<https://physicslabs.ucd.ie>

(General information, Deadlines, Slides of presentations
Python programming, Data acquisition, Data analysis)

Note: new software for 2025–2026 – please let us know if you find a broken link etc.

Email notification list for the ADVANCED PHYSICS LABORATORY

UCD-PHYSICSLABS3@listserv.heanet.ie

You should all have already been added.

Only staff can send.

Two-trimester modules

Module	Name	Credits	Laboratory Hours	Module Coordinator
PHYC30170	Astronomy and Astrophysics Lab I	20	12 hrs/wk.	John Quinn
PHYC30300	Advanced Laboratory I (Physics)	20	12 hrs/wk.	Tom McCormack
PHYC30320	Advanced Laboratory I (TP)	10	6 hrs/wk.	Luis Leon Vintro



One-trimester modules (Study Abroad)

Module	Name	Credits	Laboratory Hours	Module Coordinator
PHYC30360	Study Abroad Physics Labs (10)	10	12 hrs/wk.	Tom McCormack
PHYC30370	Study Abroad Physics Labs (5)	5	6 hrs/wk.	Tom McCormack

Study Abroad Laboratory Modules must be completed in either trimester.

Laboratory Schedule

- Trimester 1: Labs run for 12 weeks **starting today** and will close on:
 - Thursday 30th October for TP / SA (5 credit)
 - Thursday 27th November for Physics / PASS
- *Trimester 2, labs **will be open in the field –work weeks** and will close on*
 - Friday March 13th for TP / SA (5 credit)
 - Thursday 26th March for Physics / PASS
- Main Laboratory (with demonstrators):

Tues., Weds., Thurs. 2-6pm.
- You should be working on lab activities at all timetabled hours
- Sign in and out (QR code/google sheet.)



APL Staff

Staff:

- Module coordinators:**
- Prof. John Quinn (PHYC30170)
 - Dr. Tom McCormack (PHYC30300/30360/30370)
 - Assoc. Prof. León Vintró (PHYC30310/30320)

- Other staff involved:**
- Dr. Patrick Hayden
 - Assoc. Prof. Vladimir Lobaskin (Computational)
 - Dr. Morgan Fraser (Astro Experiments)
 - Assoc. Prof. Brian Vohnsen
 - Dr. Andrew Burgess

- Technical staff:**
- Mr. Paul Hanratty, Dr. Joe Branson

+ Postgraduate demonstrators



Staff



John
Quinn



Luis
León Vintró



Tom
McCormack



Paddy
Hayden



Brian
Vohnsen



Morgan
Fraser



Vladimir
Lobaskin



Andrew
Burgess



Joe
Branson

Paul
Hanratty

Number of Experiments

	Trimester 1	Trimester 2
Single Hons.	4 expts.	4 expts.
PASS	4 expts	4 expts.
TP / SA(5)	2 expts Finish in lab by 30/10/2025	2 expts Finish in lab by 13/3/2026

- A [Statement of Intent](#), signed off by a staff member, must be produced before commencing work on an experiment. (*Not required for Computational / Electronics / CCD / AIA*)
- You must submit a report before being assigned a new experiment.



Statement of Intent

UCD School of Physics
Advanced Laboratories

Name:

Experiment:

Date:

The Statement of Intent should address the following four points about the experiment you are about to undertake:

1. What is the aim of the experiment?
2. What measurements will be made and how?
3. How will the final result be obtained from the experimental data?
4. What are the main safety concerns with the experiment and precautions that should be taken?

The statement should be reviewed by a member of staff before you begin the experiment.

Motivation:

- While writing report students realise what they should have done!
- Know what you have to do in advance and make a plan!
- Safety

Equipment:

- May be different to handout (e.g AJP paper)
- You should not touch any equipment until you have discussed with a member of staff.
- It is ok (expected) to look at the equipment before writing Statement of Intent.

The Institute of Physics requested student Statements of Intent as part of the accreditation process.

- Safety in the lab is of the utmost importance.
 - People
 - Equipment
- Safety Concerns needs to be thought about in advance and included in the Statement of Intent.
- Please think critically about the safety aspects before using any equipment (ongoing).
- Equipment: we frequently have blown fuses - please know how to wire up a basic circuit including how to put an ammeter in a circuit!
- Every Stage 3 student will do the Ramsauer-Townsend Experiment that involves liquid nitrogen - everyone has to be safety trained for LN2 (next)

Progress targets for 1st trimester

	Physics / PASS / SA (10)	TP / SA (5)
Expt 1	Complete 25 th Sept	Complete 2 nd Oct
Expt 2	Complete 16 th Oct*	Complete 30th Oct Submission by 10th Nov
Expt 3	Complete 7 th Nov	NA
Expt 4	Submission by 19th Jan	NA

* Last possible **Submission 5pm on 28th Oct**

Time Management

- Managing your time effectively is critical
- Each lab is worth 2.5 ECTS credits (50-75 hrs) - 50% of a lecture module
- Indicative breakdown: assuming 3 weeks per experiment:
 - First week: familiarising with experiment: aims, equipment, set up and procedure including writing data-acquisition code (if relevant), preliminary data
 - Second week: refine experimental procedure, preliminary analysis, final data sets
 - Third week: final analysis and write report
- You will be slower at first and get faster as the year progresses!
- Do not stay stuck! Ask for help!
- Labs start today - it is vital to get into things this week.

Doing Experiments

- You must keep a laboratory notebook with a detailed record of what you did for each experiment (set-up, procedure, results).
- All data should be available (in either written or electronic form, if appropriate) if requested by staff.
- Please frequently backup all your work!
- If there is no activity on an experiment for two weeks, the experiment will be allocated to someone else.
- Experiments are designed to be extendable: if you have an idea to do something different / extra / new then please do so! (initiative and independent thinking are very important!)

- All reports are to be typed (do not forget to append Jupyter notebooks and program code to the end of your reports. Be careful with Jupyter Notebook PDF appendices that they are not all on one page!)
- An Electronic version of each report in a single pdf document is to be [submitted through Brightspace](#) (originality checker will run on it) and [a copy e-mailed to Paul Hanratty](#)
- Please remember to include an “Analysis” section in your report and do not relegate it to an undocumented Python appendix!
- Each experiment and report to be completed (submitted) before starting the next one.
- We do not want large reports – brief, factual, informative reports which are clear and understandable are preferred. (**report-writing session next week**)
- The UCD Late Submission of Coursework policy applies.

Assessment

- Single Hons. & PASS: 8 full written reports.
- TP & Joint: 4 full written reports.
- A staff member will read the report and discuss it with you. (NB. Notebook). Assessment is based on commitment, quality of work, understanding and practical skills.
- Assessment Breakdown:
 - 80% for laboratory reports and discussion with staff
 - 20% for general continuous assessment mark by the staff
- Please remember that one experiment is worth 2.5 ECTS (50-75 hrs of effort). This is 50% that of a typical lecture module and budget your time and effort accordingly!
- Grading Guide: <https://www.ucd.ie/registry/t4media/UCD%20Module%20Grade%20Descriptors-2.pdf> (link broken?)



Introductory Sessions

- *Welcome and Introduction to the Advanced Laboratories*, Prof. John Quinn, Tuesday 9 September at 14:00-14:50 (BH: B106)
- *Data Acquisition in Python*, Prof. John Quinn, Wednesday 10 September at 14:00-14:50 (BH: B106)
- *Report Writing*, Dr. Tom McCormack, Tuesday 16 September at 14:00-14:50 (BH: B106)
- *Data Analysis I*, Assoc. Prof. John Quinn, Wednesday 17 September at 14:00-14:50 (BH: B106)
- *Data Analysis II*, Assoc. Prof. John Quinn, Wednesday 24 September at 14:00-14:50 (BH: B106)
- *From Wednesday October 1st at 2pm the Student Seminars (Stage 4) start*



Dignity and Respect

- The Advanced Physics Laboratories aim to provide an educational environment with the highest standards of behaviour and ethics, free of bias and any forms of harassment or bullying.
- You should familiarise yourself with and abide by the UCD policies on Student Conduct and Academic Integrity (<https://www.ucd.ie/secca/studentconduct/>)

UCD Dignity and Respect Support Service

Supporting any student, employee or UCD community member who is impacted by issues of **bullying, harassment or sexual misconduct**

- Confidential and proactive support
- Specialist information and guidance
- Support to explore informal and formal resolution options
- Ongoing aftercare support
- Drop in or make an appointment

L532, Level 5 James Joyce Library Building

respect@ucd.ie ucd.ie/dignityandrespect

Report + Support Tool

Other Information

- A GPA of 2.48 is needed to progress to Stage 4.
- 3rd year will form 30% of the marks for the final degree.
- PASS/TP: if want to take Medical Physics module in 4th year, then you need to take Nuclear Physics module in 3rd year.
- Google Form sign in and sign out via the QR codes posted in the lab – required for attendance and safety (& insurance) reasons.
- UCD Library Physics guide: <http://libguides.ucd.ie/physics>



What now?

- Class Rep(s).
- Next: Liquid Nitrogen safety training (3pm, B000 downstairs!)
- Get started
- Have fun!